RICHMOND ESSIEKU

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EDUCATION

Texas Tech University, Lubbock, TX
Ph.D. in Economics — Expected June 2026

University of Texas Rio Grande Valley, Edinburg, TX
MSc. in Applied Statistics and Data Science — August 2022

Xiamen University, Xiamen, China
MSc. in Applied Economics (Financial Engineering) — May 2019

University of Cape Coast, Cape Coast, Ghana
BSc. in Mathematics with Economics — May 2016

FIELDS OF INTEREST

Health and LaborEconomics; Policy Evaluation; Applied Econometrics (causal inference & AI/ML)

RESEARCH

Job Market Paper

- **Diagnosed Too Soon or Too Late? Labor Market and Health Outcomes of Adults with Early vs. Late ADHD Diagnosis**, (*SSRN* Working Paper No. 5452997, 2025)
  Abstract: The study examines whether the timing of ADHD diagnosis matters. Using nationally representative U.S. Medical Expenditure Panel Survey data from 2008–2022 and a within-diagnosed design, I estimate average treatment effects with the augmented inverse probability weighting and corroborate the findings with propensity score matching, instrumental-variable estimates, and double machine learning. Earlier diagnosis delivers durable long-term health gains, raises work hours by 13.3%, and increases years of schooling, yet average labor income lags relative to later-diagnosed counterparts, with the wage penalty larger among those with more severe symptoms. A mechanism-consistent account emerges from wage equations and a Blinder–Oaxaca decomposition: Mincer marginal returns are lower for the early-diagnosed (schooling about 4.6% versus 9.4%; experience about 3.3% versus 9.2%), and the pricing of the same endowments is adverse for those diagnosed early. Timing matters: effects are strongest at ages 5–6 (work hours rise to 15.7%; health improvements largest) and remain stable through age 12. The findings motivate pairing very early screening and integrated healthcare interventions with labor-market tools that raise realized wage returns and mitigate structural barriers. Policy implications are discussed.

Publications

- "Stability analyses on the effect of vaccination and contact tracing in monkeypox virus transmission." *J. Math. Comput. Sci., 13 (2023), Article ID 8*, (Joint with Solomon Eshun and James Ladzekpo).

Working Papers

- **"In Whom We Trust? Social Networks and Trust in AI vs. Human Experts in Housing Investment Decisions"** (Joint with Prince Bosompim and Jodi Letikiewicz), Revise and Resubmit at *Finance Research Letters*.

- **"Buy Now, Struggle Later? Generational Differences in BNPL Use and Household Financial Behaviors"** (Joint with Prince Bosompim), Revise and Resubmit at *International Journal of Bank Marketing (IJBM)*.

- **"Who Benefits from Residential Solar? Evidence on Energy Insecurity Disparities and Policy Gaps in the U.S."**, Under review at *Energy Policy*.

- "When Assistance Isn't Enough: Energy Insecurity Among Older Adults and Policy Considerations for Energy Assistance Programs", Rejected and Resubmit at *Applied Economics*.

- Labor Market Margins and the Affordable Care Act's Health Insurance Reforms: Racial Heterogeneity Among Pre-Retirees, (*SSRN* Working Paper No. 5453054, 2025)
  Abstract: This study examines the causal effects of the ACA's health insurance reforms on labor market extensive and intensive margins among pre-retirees, focusing on racial disparity. Using a double robust method of propensity score matching with ATT and a staggered difference-in-differences design, we estimate the impact of Health Insurance Marketplace Subsidies (HIMS) and Medicaid expansion. Our findings indicate that HIMS delays retirement and prevents labor force exit (Number Needed to Treat, NNT, equals 132), meaning that for every 132 individuals subsidized, one additional person stays employed who would have otherwise exited. On the intensive margin, HIMS increases part-time work by 1.51 percentage points (NNT equals 66). Weekly hours drop by 1.24. Racial disparities emerge, with White and other racial groups (Asian) benefiting most, while Black and Hispanic workers experience weaker effects. No significant labor market effects are observed from Medicaid expansion.

- Mortality Risk Under Occupational and Health Comorbidity: Double Machine Learning Assessment of the Obesity–Diabetes Double Burden, (*SSRN* Working Paper No. 5453014, 2025)
  Abstract: This paper estimates the impact of type 2 diabetes on mortality among obese adults in the United States, addressing a key gap in the economics of chronic illness. Leveraging nationally representative data from the National Health Interview Survey (NHIS) linked to mortality records, we apply double machine learning to estimate the marginal impact of diabetes conditional on obesity. Results indicate that among obese individuals, diabetes increases the probability of all-cause mortality. Subgroup analyses show that this risk is especially pronounced among males, adults aged 60 to 74, individuals with hypertension, and those employed in physically demanding or public-sector occupations. A policy-relevant calculation suggests that for every 29 obese adults who avoid or delay the onset of diabetes, one premature death could be prevented. A random forest algorithm analysis highlights insurance status, educational attainment, and poverty level as dominant predictors of mortality risk, reinforcing the importance of social and economic context in shaping health outcomes.

- Predicting Obesity Risk Through Gene-Level Modeling: A Machine Learning Approach to Inform Precision Public Health Policy, (Slides # 11: ORI 8th Annual Meeting, 2023)
  Excerpt: "Can gene-level signals help us get ahead of the obesity epidemic? Leveraging DNA methylation and HPA-axis–focused feature discovery (SAM + RFE-SVM), this study builds a multilayer perceptron machine learning model that predicts individual obesity risk with strong out-of-sample performance (≈83% accuracy, 90% recall, 75% AUC). The results show how gene-based risk stratification can sharpen prevention, target resources, and inform cost-effective, precision public health policy." (Joint with Rodrigo Hansapani and Vincent Diego).

Work in Progress

- "Machine Learning Signals for Long Term Care Insurance Choice Among Older Americans" (Joint with Francis Essieku)
- "AI or Human Expert? Who Do Financial App Users Trust for Market Forecasts?"(Joint with Prince Bosompim and Gary Mottola)
- "Targeting What Works: AI Driven Clusters Inform Medication, Lifestyle, Family Context, and Workplace Supports for ADHD"
- "Multiple Job Holding and Functional Health Outcomes Across Demographic Lines"

Book Chapter

- "Machine Learning Feature Selection Techniques to Model the Elements of Cash Conversion Cycle and Other Covariates on Hospital Performance." *B.P. International, 2024*.

TEACHING

Texas Tech University

Teaching Assistant: Recitation Session and Lecture Support

- ECO 2301-001: Principles of Microeconomics — Fall 2025
- ECO 4322: The Economics of Labor Markets — Spring 2025
- ECO 2302: Principles of Economics II: Macroeconomics — Summer 2024
- ECO 3311: Intermediate Macroeconomics — Fall 2022, Spring 2024
- ECO 2305: Principles of Economics — Fall 2023
- ECO 3312: Intermediate Economic Theory — Spring 2023

University of Texas Rio Grande Valley

Instructor

- MATH 1314: College Algebra — Spring 2021
- MATH 1301: Statistics and Mathematics — Jump Start — Summer 2022

Teaching Assistant: Recitation Session and Lecture Support

- STAT 2334: Applied Statistics for Health — Fall 2021
- MATH 1460: Calculus I — Summer 2021
- MATH 2412: Pre-Calculus — Spring 2022

Regional Maritime University

Teaching Assistant: Recitation Session and Lecture Support

- BLM 105, BPS 105, DPS 105: Elements of Economics — Semester 2, 2016
- BLM 201, BPS 201, DPS 201: Principles of Economics — Semester 1, 2017
- BLM 101, BPS 101, DPS 101: Business Mathematics — Semester 1, 2020

WORK EXPERIENCE

Ghana Revenue Authority, Accra-Tema, Ghana
Statistician and Analyst — November 2015 – July 2016

- Produced high-frequency statistical reports on revenue mobilization and public finance operations, applying economic reasoning to inform tax policy and compliance monitoring.
- Conducted data-driven evaluations of taxpayer behavior and revenue trends contributing to internal auditing processes and econometric insights for decision support.

ACADEMIC SERVICES

Academic Journal Reviewer — 2023 – 2025

- Completed 15 peer reviews for the following journals: *International Economics and Economic Policy, Journal of Data Analysis and Information Processing (JDAIP), Journal of Service Science and Management, iBusiness*.

Conference Abstract Reviewer — 2024 – 2025

- Reviewed abstract submissions for academic conferences including: Southwest Decision Science Institute (SWDSI) Annual Conference (2025), IntelliSys Conference (2024–2025), IACMR Conference (2025).

Academic Judge – Student Research and Presentation Competitions — 2023 – 2025

- Served as judge for various competitions at Texas Tech University, including: 3-Minute Thesis Competition (2023–2024), Art and Humanities Conference (2023–2024), Undergraduate Research Competition (2025), Graduate Student Research Poster Competition – Abstract Committee (2024).

CONFERENCES AND PRESENTATIONS

- Southern Economic Association 95th Annual Meeting — 2025 (scheduled)
- The Washington DC Conference on the Social Sciences — 2026 (scheduled)
- Administrative Data Research Conference — 2025
- International Forum on Research Excellence, Sigma Xi — 2025
- World Obesity and Weight Management Congress — 2024
- Obesity Research Institute Annual Conference — 2023, 2024
- College of Sciences Research Conference, UTRGV — 2024
- West Texas Health Disparities Symposium, TTU — 2024
- Graduate Student Research Poster Competition, TTU — 2023, 2024

SCHOLARSHIPS AND AWARDS

Academic Scholarships and Fellowships

- Helen Devitt Jones Fellowship Grant, Texas Tech University — 2022–Present
- Graduate Assistantship, Department of Economics, Texas Tech University — 2022–Present
- Graduate Assistantship, Department of Mathematics and Statistics, UTRGV — 2021–2022
- The Washington DC Conference on the Social Sciences academic travel grant — 2025
- Department of Economics - Texas Tech University academic travel grant — 2025
- Chinese Government Council Scholarship — 2017–2019
- HEERF Grant, State of Texas — 2021

Research and Competition Awards

- First Place, 22nd Texas Tech Graduate Student Research Poster Presentation — 2023
- First Place, Kaggle Data Mining Competition (Team won over 11 competing teams) — 2022

PROFESSIONAL MEMBERSHIP

- American Economic Association (AEA)
- American Society of Health Economists (ASHEcon)
- Southern Economic Association (SEA)
- Sigma Xi – The Scientific Research Honor Society

COMPUTER SKILLS

• R • Python • SAS • STATA • LaTeX

REFERENCES

Professor Kaj Gittings
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Texas Tech University
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